ABHIJAY GANGARAPU

Independent Researcher

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EDUCATION

McNeil High School Austin, TX
Expected Graduation: May 2030 Incoming Freshman
Two grades ahead in mathematics

RESEARCH INTERESTS

Arithmetic dynamics, p-adic analysis, neural network limitations, mechanistic interpretability, theoretical machine learning, ergodic theory, pseudorandomness

RESEARCH EXPERIENCE

Neural Cryptanalysis: Period Growth in Piecewise Affine Systems 2025–Present
Independent Research Project

- Proved super-linear period growth theorem for piecewise affine maps over $\mathbb{Z}/p^k\mathbb{Z}$ satisfying Hensel condition: $T(p^{k+1}) > p \cdot T(p^k)$
- Validated across 168 primes with mean growth ratio 26.94× (max 79.42×)
- Characterized neural resistance threshold at $T/L_{\text{in}} \approx 21$ where Transformers collapse from 100% to $< 3\%$ accuracy
- Established measure-preserving property on p-adic unit ball with ergodic interpretation
- Derived theoretical formula $T_{\text{crit}} \propto N_{\text{train}}/\log m$ connecting to VC dimension and Rademacher complexity
- Proposed p-adic positional encodings as architectural fix for Transformer blindness to hierarchical structure
- 100% reproducible codebase (24/24 verification checks pass)
- github.com/CoderAwesomeAbhi/neural-cryptanalysis

TECHNICAL SKILLS

Programming: Python (NumPy, PyTorch, scikit-learn), LaTeX, Git
Mathematics: Number theory, p-adic analysis, dynamical systems, measure theory, linear algebra, abstract algebra
Machine Learning: Neural network architectures (MLP, LSTM, Transformer), training pipelines, mechanistic interpretability, statistical analysis
Tools: Berlekamp–Massey algorithm, cycle detection (Floyd's, Brent's), bootstrap confidence intervals, hypothesis testing

PUBLICATIONS & PRESENTATIONS

Period Growth and Neural Predictability in Piecewise Affine Systems over Residue Rings
Research Paper (2026)
Hensel lifting, cross-prime validation, and an empirical neural resistance threshold. 26-page paper with complete proofs, p-adic measure theory, and theoretical derivation of neural threshold.

AWARDS & HONORS

Preparing for ISEF 2027 submission

RELEVANT COURSEWORK

Abstract Algebra, Number Theory, Real Analysis, Linear Algebra, Probability & Statistics, Multivariable Calculus

ADDITIONAL INFORMATION

Research Philosophy: Rigorous mathematical proofs combined with comprehensive empirical validation. All results fully reproducible with open-source code.

Interests: Exploring fundamental limitations of AI systems on structured mathematical problems, connecting discrete dynamics to continuous measure theory, understanding when and why neural networks fail.